BUG REPORT SUMMARY - INTERNOVA PROJECT (SPRINT 1 QA) IT23631960 – H M H K HERATH

Document Date: March 2, 2026
QA Phase: Test Execution & Analysis
Total Bugs Found: 5
Critical (P0): 2 | High (P1):** 3

BUG SUMMARY TABLE

Bug ID	Title	Severity	Priority	Component	Test Case
BUG-001	Email Format Validation Not Enforced	HIGH	P1	Backend API	TC-AUTH-003
BUG-002	Weak Password Validation Missing	HIGH	P1	Backend API	TC-AUTH-004
BUG-003	SQL Injection Vulnerability	CRITICAL	P0	Backend Validation	TC-AUTH-006
BUG-004	XSS Vulnerability in Input	CRITICAL	P0	Backend Input	TC-AUTH-007
BUG-005	Rate Limiting Not Implemented	HIGH	P1	Backend Auth	TC-AUTH-013

DETAILED BUG REPORTS

BUG-001: Email Format Validation Not Enforced

Severity: HIGH | **Priority:** P1
Component: POST /api/auth/register
Test Case: TC-AUTH-003

Issue: Invalid emails (@example.com, test@, invalid-email) accepted instead of rejected.

Steps to Reproduce:

POST /api/auth/register

```
{
  "email": "@example.com",
  "password": "SecurePass123!",
  "fullName": "Test",
  "role": "Student"
}
```

Expected: 400 Bad Request
Actual: 201 Created (email accepted)

Root Cause: RegisterRequest uses only [EmailAddress] attribute without strict regex validation.

BUG-002: Weak Password Validation Missing

Severity: HIGH | **Priority:** P1
Component: POST /api/auth/register
Test Case: TC-AUTH-004

Issue: Accepts weak passwords: “weak”, “123”, single characters. No complexity requirements.

Steps to Reproduce:

POST /api/auth/register

```
{
  "password": "weak",
  "email": "test@example.com",
  "fullName": "Test",
  "role": "Student"
}
```

Expected: 400 Bad Request (weak password)
Actual: 201 Created

Root Cause: No password strength validation in RegisterRequest DTO.

BUG-003: SQL Injection Vulnerability

Severity: CRITICAL | **Priority:** P0 (FIX IMMEDIATELY)
Component: POST /api/auth/register
Test Case: TC-AUTH-006

Issue: SQL injection patterns accepted in email field (defense-in-depth failure).

Steps to Reproduce:

POST /api/auth/register

```
{
  "email": "admin' OR '1'='1@test.com",
  "password": "SecurePass123!",
  "fullName": "Attacker",
  "role": "Student"
}
```

Expected: 400 Bad Request (injection rejected)
Actual: 201 Created (pattern stored in database)

Root Cause: Minimal input validation allows dangerous characters ('", ;, –, /, /) in data.

BUG-004: XSS Vulnerability in User Input

Severity: CRITICAL | **Priority:** P0 (FIX IMMEDIATELY)
Component: POST /api/auth/register
Test Case: TC-AUTH-007

Issue: Full Name field accepts HTML/JavaScript for potential XSS attacks.

Steps to Reproduce:

POST /api/auth/register

```
{
  "fullName": "<script>alert('XSS')</script>",
  "email": "test@example.com",
  "password": "SecurePass123!",
```

```
  "role": "Student"
}
```

Expected: 400 Bad Request (HTML rejected)
Actual: 201 Created (script stored in database)

Root Cause: No HTML sanitization or character validation on Full Name field.

BUG-005: Rate Limiting Not Implemented

Severity: HIGH | **Priority:** P1
Component: POST /api/auth/login
Test Case: TC-AUTH-013

Issue: Unlimited login attempts allowed. No brute force attack protection.

Steps to Reproduce:

1. Send 50+ failed login requests in 1 minute
2. Expected: 429 Too Many Requests after 5 attempts
3. Actual: All requests processed (no rate limiting)

Root Cause: No rate limiting middleware configured on login endpoint.

Impact: Attackers can brute force user passwords without throttling.

SEVERITY & PRIORITY MATRIX

Severity Levels

Level	Meaning	Release?
CRITICAL	Data breach, system failure	❌ NO
HIGH	Major feature broken, security risk	⚠️ Fix first
MEDIUM	Degradation, minor issue	Later
LOW	Cosmetic, nice-to-have	Backlog

Priority Levels

Priority	Timeline	Release Impact
P0	Immediate	Blocking production
P1	This sprint	Major feature
P2	Next sprint	Can schedule
P3	Backlog	Future

TEST EXECUTION RESULTS

Framework	Tests	Passed	Failed	Status
xUnit (Backend)	14	8 ✅	6	Complete
Jest (Frontend)	13	6 ✅	7	Complete
Total	**27**	**14**	**13**	**Done**

✅ Failed tests prove bugs exist - test automation working correctly.

SPRINT 1 RECOMMENDATIONS

Must Fix Before Release (P0)

- BUG-003: SQL Injection
- BUG-004: XSS Protection

Fix This Sprint (P1)

- BUG-001: Email validation
- BUG-002: Password strength
- BUG-005: Rate limiting

Re-test After Fixes

dotnet test Internova.Tests
npm test

QA Status: TESTING COMPLETE ✅
Date: March 2, 2026
Version: 1.0